Rishabh Shah
Junior Undergraduate
Discipline of Mechanical Engineering
Indian Institute of Technology, Gandhinagar

24110327@iitgn.ac.in
9408373590
Website | LinkedIn | Github

ACADEMIC DETAILS

Degree	Specialization	Institute	Year	CPI/%
B.Tech.	*Mechanical Engineering*	IIT Gandhinagar	2024-2028	8.30
Class XII	*Physics, Chemistry, Maths*	Cosmos Castle International School	2023-2024	88.89%
Class X		Navkar Public School	2021-2022	93.5%

INDUSTRIAL INTERNSHIP

- **Robotics Engineering Intern (Localization and Sensor Fusion)** [January 2026-August 2026]
  (Advisor:- Prof. Madhu Vadali | Krishna Defence | Research & Development | Robotics Lab, Gandhinagar, Gujarat) | Project Link
  - Developed a **ROS 2** localization system to enable forklift (mobile robot) **Navigation** in GPS-denied warehouse environments
  - Mounted **Ultra-Wideband** (UWB) and **IMU** sensors on a mobile robot to capture real-time data
  - Developed **sensor fusion algorithms (EKF)** to actively filter UWB signal noise and correct continuous IMU hardware drift
  - Achieved highly precise real-time state estimation, resulting in a reliable indoor tracking **accuracy of 9.7 cm** by verifying with Robotic Total Station

PROJECTS

- **Pick and Place of Objects for Heal & Franka Manipulators** [September 2026 - October 2026]
  (Advisor:- Prof. Madhu Vadali) | Project Link
  - Built a MuJoCo simulation to automate pick-and-place tasks for Addverb HEAL and Franka Panda robotic arms, implementing a 7-phase state machine with grasp verification.
  - Developed and compared three distinct Inverse Kinematics (IK) solvers–Mink, Damped Least Squares (DLS), and Quadratic Programming (QP), to optimize the robot's trajectory planning and joint movement.
- **Autonomous UR5e Control Simulation (MuJoCo)** [December 2025]
  Project Link
  - Simulated a **6-DOF UR5e manipulator** in MuJoCo by mapping the complete kinematic chain and **physical joint** constraints
  - Implemented rigid body dynamics and **Inverse Kinematics (IK)** for precise end-effector positioning, establishing a baseline Reinforcement Learning using Python and Stable-Baselines3
- **Design and Kinematic Analysis of a Crank-Rocker Mechanism** [October 2025 - November 2025]
  (Advisor:- Prof. Jayaprakash) | Project Link
  - Designed and fabricated a low-friction planar four-bar **Crank-Rocker mechanism**, and coded a MATLAB script to automate theoretical position and velocity analysis using **Grashof's Law** and vector loop equations
  - Validated the theoretical model through experimental video tracking, engineering a Python script to process position data, apply numerical differentiation, and perform comprehensive error analysis

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++
- **Libraries:** Pandas, Numpy, Matplotlib
- **Tools:** Autodesk, Ansys (Basic), GitHub, MSC Adams (Basic), MATLAB, Mathematica
- **Robotics:** ROS 2, MuJoCo

POSITIONS OF RESPONSIBILITY

- **Coordinator, Tinkerer's Lab** [May 2026 - Present]
  - Optimized lab infrastructure by streamlining the laser-cutting booking portal and leading the wireless network integration for 3D printers
  - Directed hands-on manufacturing workshops for 150 students and authored standardized equipment manuals to accelerate the technical onboarding of freshers
- **Sponsorship Team Senior Member, Blithchron IITGN** [August 2025 - February 2026]
  - Completed training in corporate outreach to understand how to strategically approach companies for sponsorships.
  - Conducted mock interview sessions to practice and refine professional communication and pitching skills

ACHIEVEMENTS AND EXTRACURRICULAR ACTIVITIES

- Awarded Academic Citations for Academic Excellence in the 4th Semester
- Secured All India Rank (AIR) 3630 in JEE Mains and AIR 5433 in JEE Advanced
- Secured 1st place among 50+ teams in the Hallabol Chess Tournament at IIT Gandhinagar as a member of the winning team
- Secured the Gold Medal and Championship Trophy at the KnightPlayers Grand Chess Championship, IITGN; played in all tournament fixtures and was awarded Man of the Match in the finals